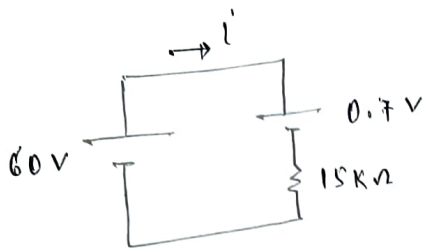
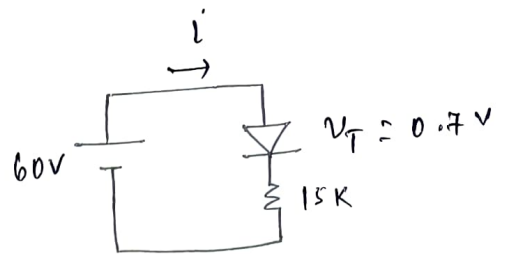


1. (a) Assume that Diode is forward biased then the circuit will become

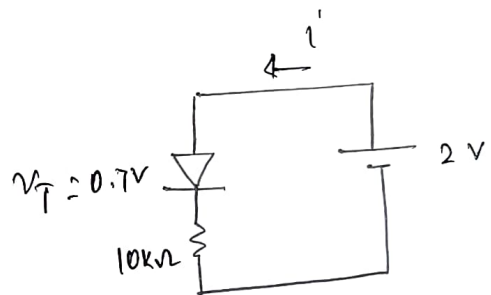


$$\Rightarrow i' = \frac{60 - 0.7}{15 \text{ K}\Omega} = 3.95 \text{ mA}$$

As the current value is +ve and flowing through anode to cathode, so, our assumption is correct, and calculated value of i' is the final answer.

- (b) Assume diode is F.B.

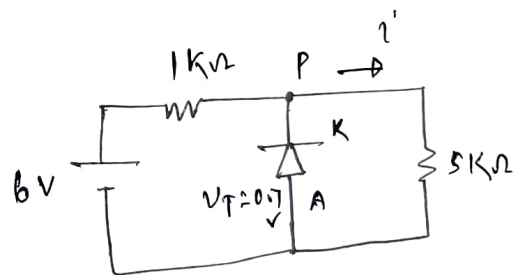
$$i' = \frac{2 - 0.7}{10 \text{ K}\Omega} = 0.13 \text{ mA}$$



and $i' > 0$; so assumption is correct.

- (c) Assume diode is Reversed biased

$$\text{so, } i' = \frac{6 \text{ V}}{6 \text{ K}\Omega} = 1 \text{ mA}$$



$$V_P = 6 - 1 = 5 \text{ V}$$

$$V_{AK} = 0 - 5 = -5 \text{ V} \quad \text{i.e. less than } V_T = 0.7 \text{ V}$$

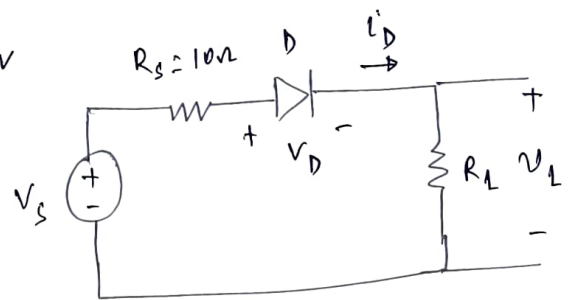
so, our assumption is correct and calculated answer is the final answer.

② Assume diode is ideal i.e. $V_T = 0V$

then KVL:-

$$V_s = R_s i_D + V_L$$

so, $V_L = V_s - R_s i_D$



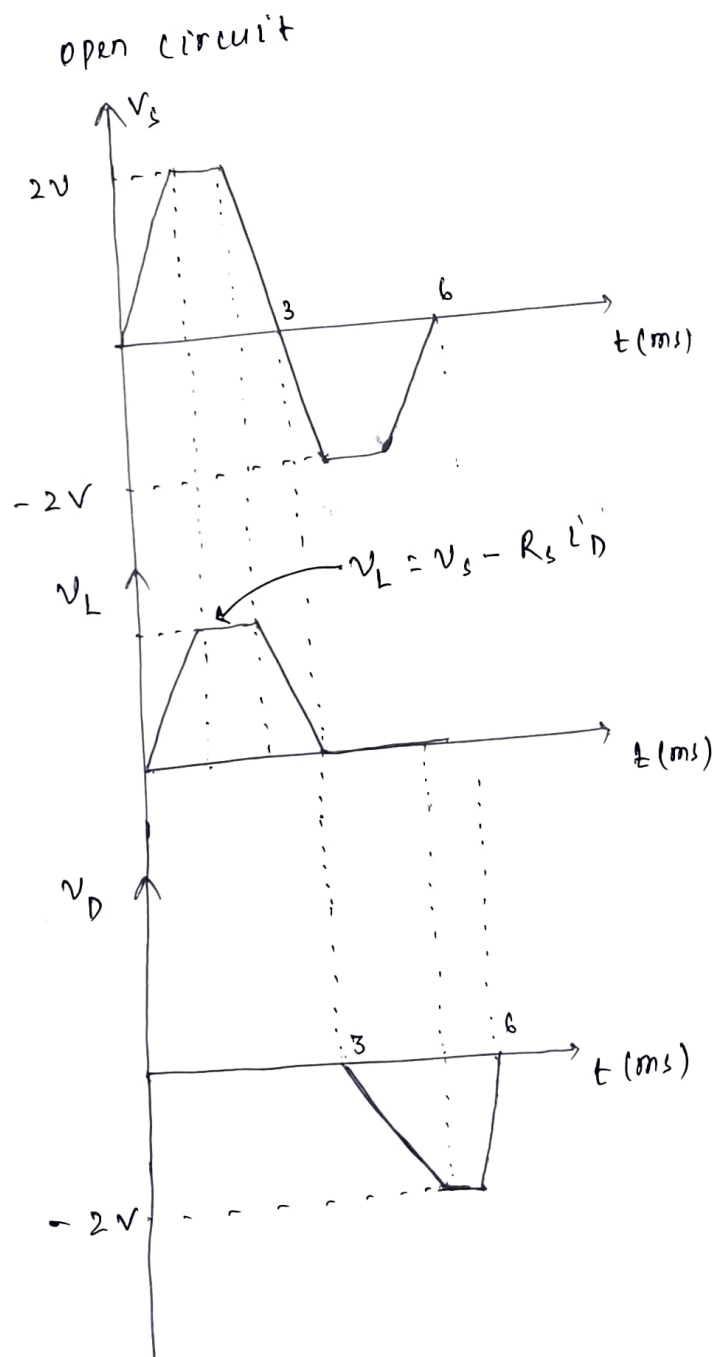
When $V_s > 0$ (Diode will be F.B.)

When $V_s < 0$

Diode will be R.B. $\Rightarrow$ open circuit

so, $V_L = 0V$

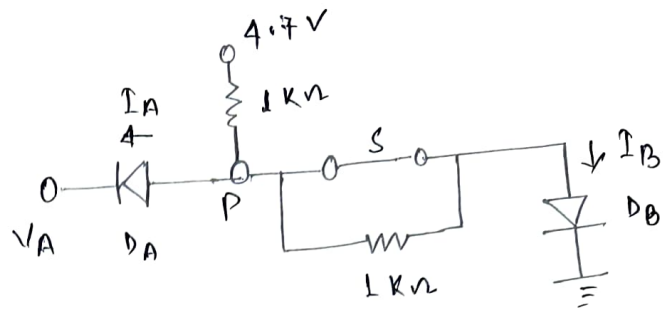
KVL:- $V_D = V_s$



3) a)

Given $V_t = 0.3 \text{ V}$

for $V_A = -1 \text{ V}$



Assume Diode A (D_A) is on & D_B is off

KVL :- $4.7 = 1000 I_A + 0.3 - 1$

so, $I_A = 5.4 \text{ mA}$ $\rightarrow$ Assumption was correct.

$V_P = (4.7 - 5.4) \text{ V} = -0.7 \text{ V}$

$V_P < V_t$; so D_B is off $\rightarrow$ Assumption is correct

so, $I_B = 0$

For $V_A = -0.1 \text{ V}$

Assume D_A is ON & D_B is off.

KVL :- $4.7 = 1000 I_A + 0.3 - 0.1$

$I_A = 4.5 \text{ mA}$ $\rightarrow$ Assumption was correct.

$V_P = (4.7 - 4.5) \text{ V} = 0.2 \text{ V}$

so, $V_P < V_t$; so, D_B is off $\rightarrow$ Assumption was correct.

$I_B = 0$

for $V_A = 1V$

Assume D_B is ON, D_A is off

KVL :- $4.7 = 1000 I_B + 0.3$

$I_B = 4.4 \text{ mA}$ $\rightarrow$ Assumption was correct.

$V_p = (4.7 - 4.4) \text{ V} = 0.3 \text{ V}$

$V_{PA} = 0.3 - 1 = -0.7 \text{ V}$

so, $V_{PA} < V_t \Rightarrow D_A$ is off $\rightarrow$ Assumption was correct.

$I_A = 0$

for $V_A = 2V$

Assume D_B is ON, D_A is off

KVL :- $4.7 = 1000 I_B + 0.3$

$I_B = 4.4 \text{ mA}$ $\rightarrow$ Assumption was correct.

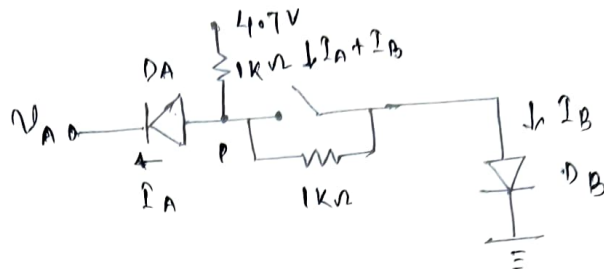
$V_p = (4.7 - 4.4) \text{ V} = 0.3 \text{ V}$

$V_{PA} = -1.7 \text{ V}$

$V_{PA} < V_t \Rightarrow D_A$ is off $\rightarrow$ Assumption was correct.

$I_A = 0$

3. (b) When switch S is open & $V_A = 1V$, $V_t = 0.3V$



Assume D_B is ON & D_A is ~~OFF~~ also ON

So, KVL (Right side of circuitry) :-

$$4.7 = 1000(I_A + I_B) + 1000I_B + 0.3$$

$$\Rightarrow 4.4 = 1000I_A + 2000I_B \quad \text{--- (1)}$$

KVL (Left side of circuitry) :-

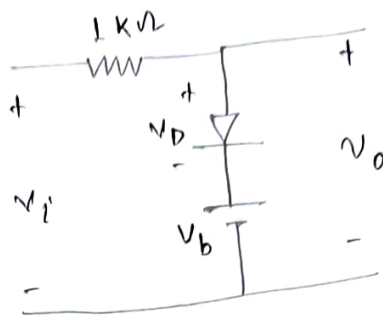
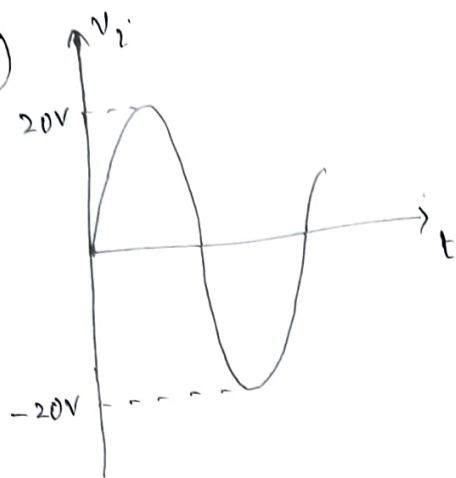
$$4.7 = 1000(I_A + I_B) + 0.3 + 1$$

$$3.4 = 1000(I_A + I_B) \quad \text{--- (2)}$$

Solving eqⁿ (A) & (B), we get $I_A = 2.4mA$ & $I_B = 1mA$

Since +ve value of I_A, I_B makes D_A, D_B ON & hence our assumption becomes correct. Hence, our calculated values can be taken as final value.

Q4



Let diode is ideal ; so, $V_f = 0V$

$V_b = 10V$ (Given)

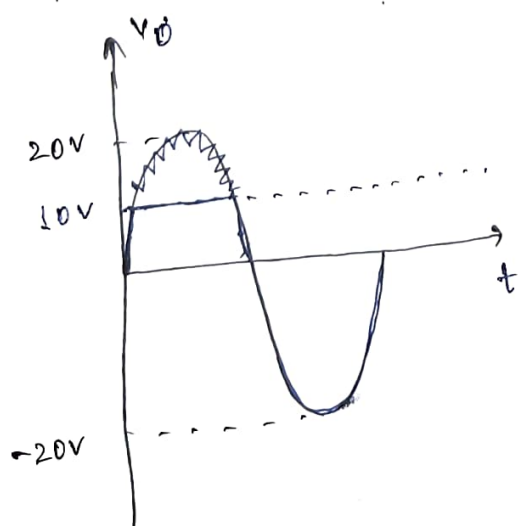
Diode will get ON, when $V_i \geq V_b$ i.e. $V_i \geq 10V$

When the diode is ON, it will be short circuited and

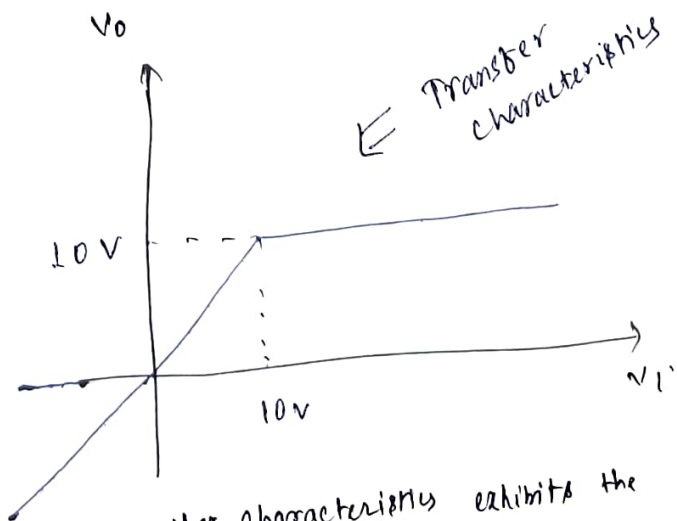
$V_o = V_b = 10V$

Diode will be OFF, when $V_i < V_b$ i.e. $V_i < 10V$

so, no current flows in the circuit & $V_o = V_i$

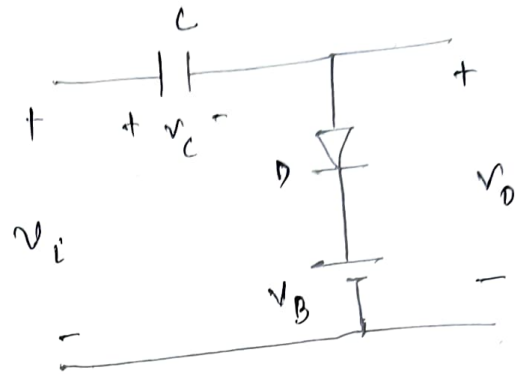
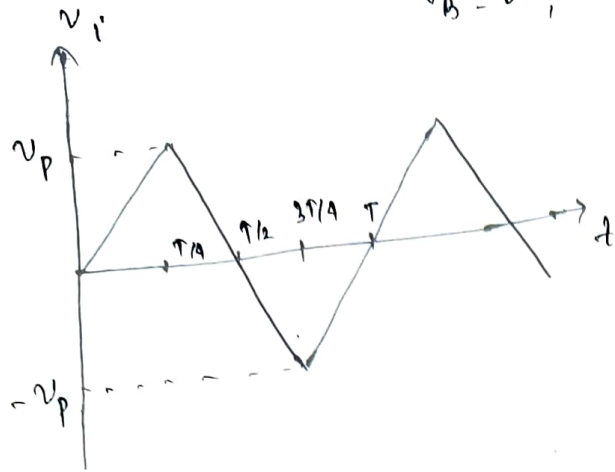


← Output waveform for one cycle



Transfer characteristics exhibits the relation between input & output.

⑤ Given, C is initially uncharged; i.e. $V_C = 0V$
 $V_B = 0$, & assume diode is ideal; i.e. $V_T = 0V$



so, from $0 \leq t < T/4$

$V_C = 0$, diode will be forward biased & $V_o = V_D$ as
 $V_B = 0$. so, diode will be short-circuited and

$$V_o = V_B = 0V$$

Also, KVL:- $V_C = V_i - V_o$
 $V_C = V_i$

& at $t = T/4$; $V_C = V_p$

capacitor is charged by V_p voltage.

From $T/4 \leq t < T/2$

$$V_D = V_{\text{Anode}} - V_{\text{Cathode}}$$

$$V_{\text{Anode}} = V_C = V_p$$

$$V_{\text{Cathode}} = 0V$$

& $V_{\text{input}} < V_p$ (so, diode will be off)

so, $V_o = V_i - V_C$ — (By KVL)

Note:- Capacitor has no path to discharge.

at $t = T/2$

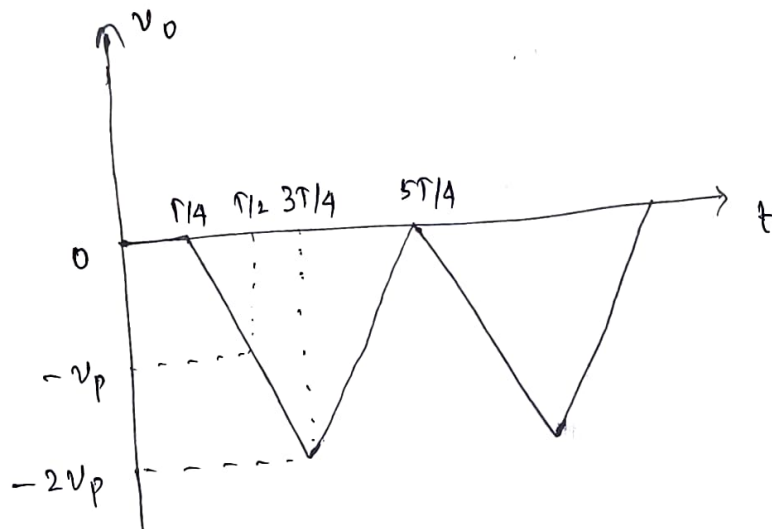
$v_i = 0, \quad v_c = v_p$

so, (A) $\Rightarrow \boxed{v_o = -v_p}$

at $t = \frac{3T}{4}; \Rightarrow v_i = -v_p$

(A) $\Rightarrow \boxed{v_o = -2v_p}$

Similarly, it will follow the KVL based on input voltage.



⑥ Given

$$R = 100 \Omega$$

$$C = 20 \mu\text{F}$$

Diode is ideal (let)

capacitor is initially uncharged

Now, $0 \leq t < 2\text{ms}$

Diode will be forward biased
circuit will close, current will
flow & capacitor will start
charging.

$$V_c(t) = V_c(\infty) + [V_c(0) - V_c(\infty)] e^{-t/\tau} \quad \text{--- (A)}$$

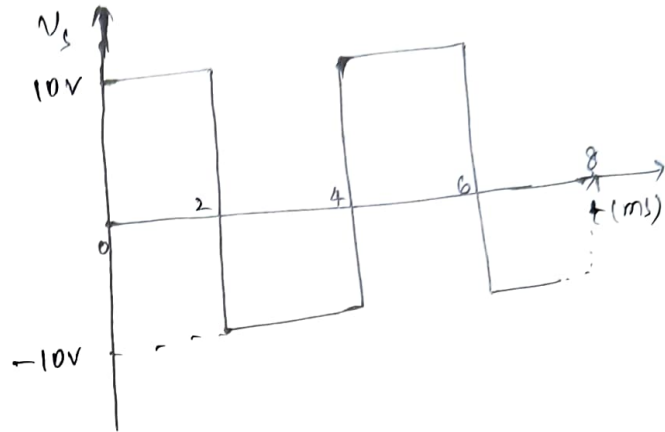
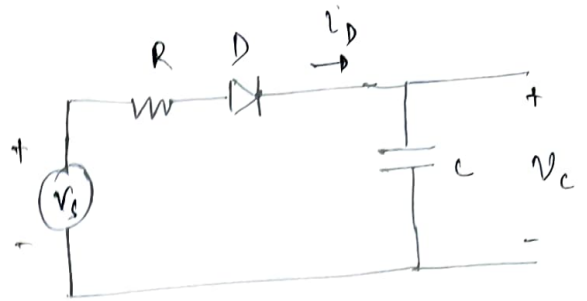
$$\tau = RC = 100 \times 20 \times 10^{-6} = 2\text{ms}$$

⑥ ~~A~~ $\rightarrow$ so, at $t = 2\text{ms}$.

$$\textcircled{A} \Rightarrow V_c(t) = 6.32\text{V}$$

from $2\text{ms} \leq t < 4$

Now, Diode will be Reverse biased & circuit will open. There
will be no path to discharge the capacitor. so, capacitor will
hold its voltage value.



From $4\text{ms} \leq t < 6\text{ms}$

Diode will be forward biased and once again capacitor will start charging from 6.32V

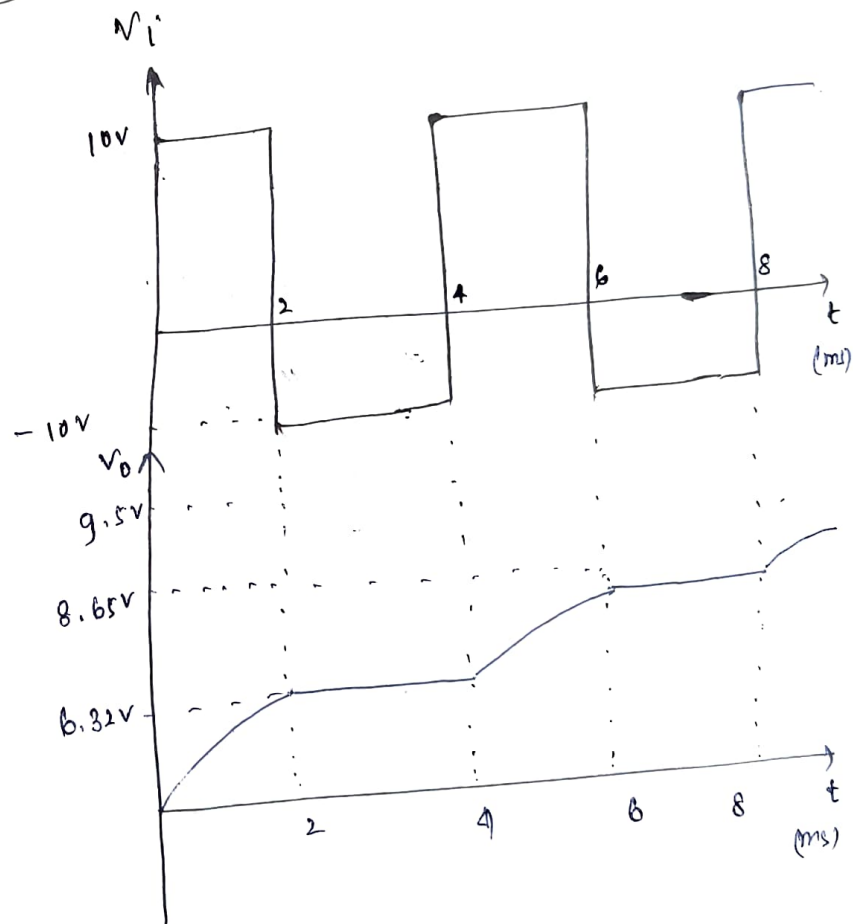
$$\text{so, } v_c(t) = v_c(\infty) + [v_c(0) - v_c(\infty)] e^{-(t-4)/\tau} \quad \text{--- (B)}$$

Here $v_c(0) = v_c(2) = 6.32\text{V}$

so, at $t = 6\text{ms}$

$$\text{(B)} \Rightarrow v_c(6) = 8.65\text{V}$$

By, $v_c(10) = 9.5\text{V}$



7

Given diodes are ideal

so, $V_f = 0V$

For $-20 \leq V_1 \leq 20 V$

$V_2 = ?$

so, when $-20 \leq V_1 < 0 V$
 case 1

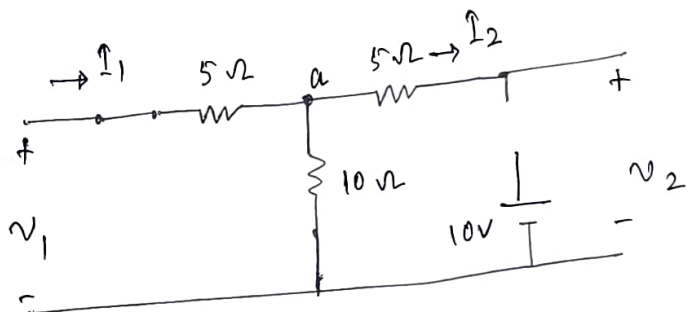
D_1 & D_2 both will be off & hence $V_2 = 0 V$

case 2 when $0 \leq V_1 \leq 15 V$

Assume D_1 is ON & D_2 is off

apply nodal analysis at a

$$\frac{V_a}{10} + \frac{V_a - V_2}{5} + \frac{V_a - V_1}{5} = 0 \quad \text{--- (1)}$$



& $V_2 = V_a$

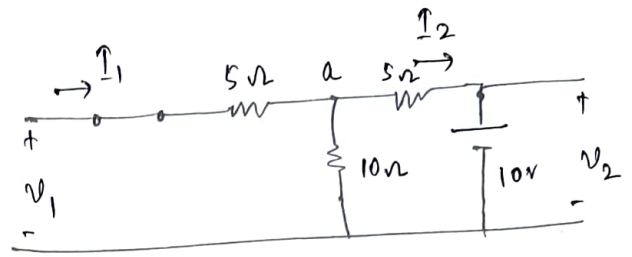
so, (1) $\Rightarrow V_a = \frac{2}{3} V_f \Rightarrow V_2 = V_a = \frac{2}{3} V_f$

so, if $V_f = 12 V \Rightarrow V_a = 8 V$ & D_A will be ON, D_B will be off $\Rightarrow$ Assumption was correct.

case 3 when $15 < V_1 \leq 20$ V

Assume D_1 & D_2 both are ON

Now, $V_2 = 10$ V



apply nodal at a

$$\frac{V_a - V_1}{5} + \frac{V_a}{10} + \frac{V_a - 10}{5} = 0$$

$$V_a = 4 + 0.4 V_1$$

if we take $V_1 = 16$ V, then $V_a = 10.4$ V & hence D_1 & D_2 both will ON $\Rightarrow$ so our assumption was correct.

so, finally,

$$V_2 = \begin{cases} 0, & -20 \leq V_1 < 0 \text{ V} \\ \frac{2}{3} V_1, & 0 \leq V_1 \leq 15 \text{ V} \\ 10 \text{ volt} & \text{when } 15 < V_1 \leq 20 \end{cases} \quad (\text{volt})$$

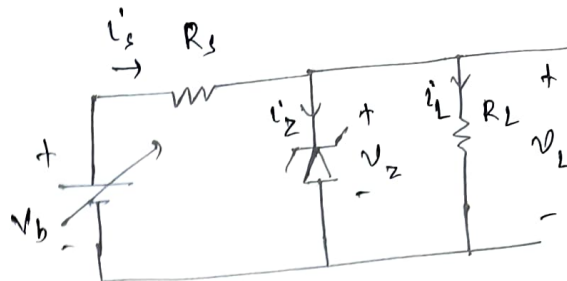
(8) Given

$$R_L = 9 \Omega$$

$$V_Z = 8.2 \text{ V} \quad \text{and} \quad 75 \text{ mA} \leq I_Z \leq 1 \text{ A}$$

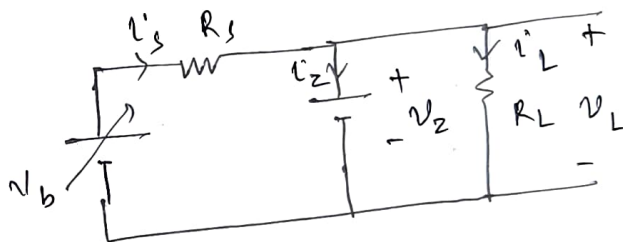
$$V_L = 8.2 \text{ V (regulated)}$$

$$V_b = 12 \text{ V } (\pm 10\%)$$



So, A/q, when the output voltage is regulated at 8.2 V then the Zener diode will allow the flow of current through it in the range of $[75 \text{ mA}, 1 \text{ A}]$

$$\text{So, } I_L = \frac{V_L}{R_L} = \frac{8.2}{9} \text{ A} = 0.91 \text{ A}$$



$$I_Z = I_s - I_L \quad \text{--- (A)}$$

$$I_s = \frac{V_b - V_Z}{R_s} \quad \text{--- (B)}$$

$$\text{So, (A) } \Rightarrow I_Z = \frac{V_b - V_Z}{R_s} - 0.91 \quad \text{--- (C)}$$

$$\text{Now, } 0.075 \leq I_Z \leq 1 \text{ A} \quad \text{--- (D)}$$

$$\text{and } 10.8 \leq V_b \leq 13.2 \text{ V} \quad \text{--- (E)}$$

So, through care! when $I_Z > 0.075 \text{ A}$

$$(c) \Rightarrow \frac{V_b - V_Z}{R_s} = 0.91 > 0.075$$

When $V_b = 10.8 \text{ V}$

$$(c) \Rightarrow \frac{10.8 - 8.2}{R_s} = 0.91 > 0.075$$

$$R_s \leq 2.64 \Omega$$

(1)

* When $V_b = 13.2 \text{ V}$

$$(c) \Rightarrow \frac{13.2 - 8.2}{R_s} = 0.91 > 0.075$$

$$R_s \leq 5.07 \Omega$$

(2)

so, from (1) & (2) $\Rightarrow R_s \leq 2.64 \Omega$ — (A1)

Case 2

When $I_Z \leq 1 \text{ A}$

$$(c) \Rightarrow \frac{V_b - V_Z}{R_s} = 0.91 \leq 1$$

When $V_b = 10.8 \text{ V}$

$$(c) \Rightarrow R_s \geq 1.36 \Omega$$

(3)

* When $V_b = 13.2 \text{ V}$

$$(c) \Rightarrow R_s \geq 2.62 \Omega$$

(4)

so, from (3) & (4)

$$R_s \geq 2.62 \Omega$$

(A2)

so, from (A1) & (A2)

R_s should lie in between
 2.62Ω to 2.64Ω .